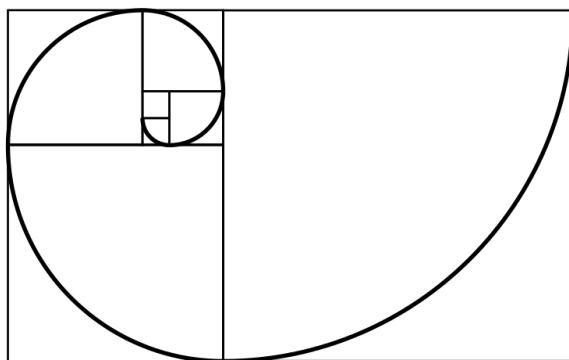


Quantum Limited Gravitational Wave Detection

University of Glasgow



Joseph Henry Briggs

Day Month Year

Abstract

All you need are lasers and photodiodes.

Dedication

Declaration

I declare that..

Acknowledgements

I want to thank...

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Chapter 1

Gravitational Wave Interferometers

This chapters purpose is to link general relativity to a basic michelson interferometer. I will keep the general relativity to a minimum, since this isn't a textbook on GR, however there is a small amount required to know why gravitational wave detectors are so big, why they are L shaped, etc. It will then expand on the basic michelson by adding the cavities. This is a good point to link sources to the detector, since the signal recycling adds shaping, its good to know why we aimed for 100Hz (I actually don't know why this was done, for some reason LIGO is optimised for detecting heavy black holes I think). Talk about noise sources, noise and sensivity curves, transfer functions between differential arm motion to light modulation at the AS port, mention the standard quantum limit. A detailed explanation of shot noise and radiation pressure noise should be in here, since these basically give you the requirements for the laser chapter and the photodiode chapter. Once the noise sources are covered, the BNS range can be shown. It makes sense to introduce the current detectors in more detail, and then talk about the 3rd generation of detectors. More novel topologies can be mentioned after.

1.1 Introduction

1.2 What is a gravitational wave

1.3 Little bit of astrophysics and cosmology

1.3.1 Sources of gravitational waves

formula for neutron star distance is the only important thing here, since this drives the low losses

- 1.3.2 How detections of gravitational waves have verified GR
- 1.3.3 how gravitational wave detections have been important to cosmology
- 1.3.4 how gravitational wave detections have been important to astronomy
- 1.4 Section linking GR to interferometer with light like spacetime intervals
- 1.5 Amplitude and Phase modulated light
- 1.6 Fabry-Pérot Cavity as a Gravitational Wave interferometer
- 1.7 Basic Michelson interferometer
 - 1.7.1 dark fringe
- 1.8 Michelson with arm cavities
- 1.9 Michelson with power recycling
- 1.10 Michelson with signal recycling
- 1.11 Response of dual Michelson to a gravitation wave
- 1.12 Sensitivity curves
- 1.13 Noise sources
 - 1.13.1 Shot noise
 - 1.13.2 All noise sources...
- 1.14 losses

Show how losses = less range

1.15 Third Generation Ground based Detectors

1.16 other detectors

1.16.1 LISA

1.16.2 speedmeter

1.17 Gaussian beams

Chapter 2

High Power Laser Prototype for LIGO

2.1 Introduction

- why ligo needs high power
- brief history of LIGO laser
- mention noise problems leading to wanting new amplifier

2.2 Requirements

- power requirements
- long term stability requirement
- noise requirements (pointing+frequency+amplitude, only measured frequency and amplitude)

2.3 OEM solid-state Amplifier

- why are they better than an HPO in principle
- how they work
- neovan 4S-HP

2.4 Set-up and Layout

- diagram of layout
- description of layout
- use 2 amplifiers and a PMC to get high power gaussian beam

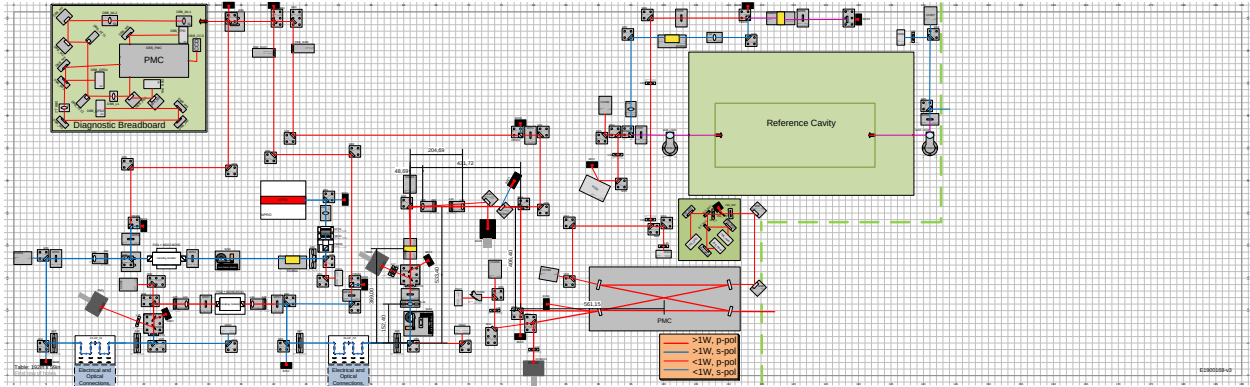


Figure 2.1: e1900168. Layout placeholder

- flow rates and beam quality of amplifiers

2.5 Frequency Noise

Need to put together the measurements for frequency noise.

2.6 Long Term Power Results

In touch with Matt about his, I would like 1 month or something like that (as long as possible)

2.7 Conclusion

summary of how much power for how long, and how it could be improved slightly with mode matching

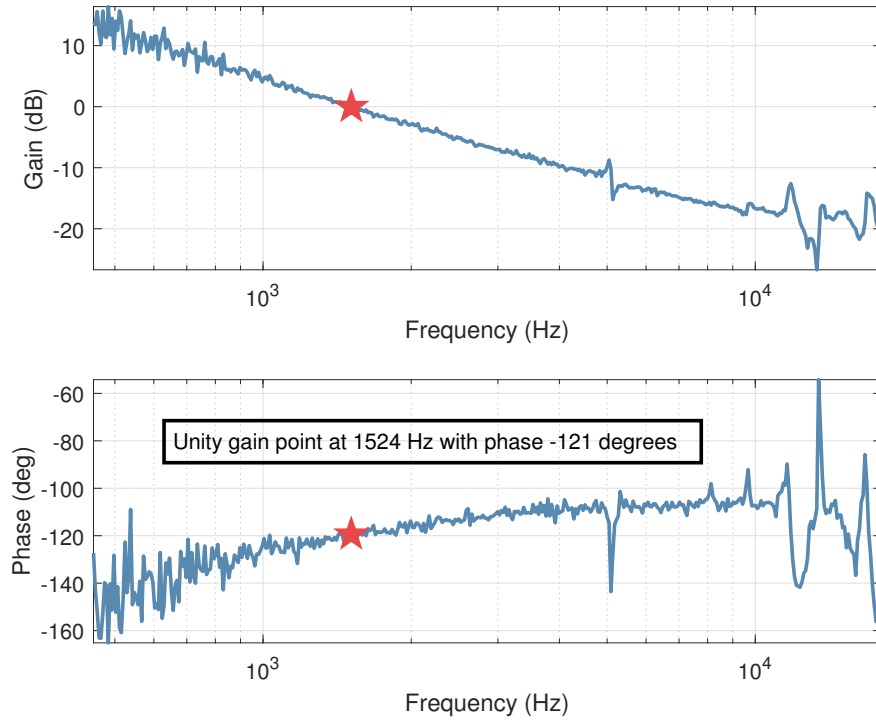


Figure 2.2: Open loop gain of frequency stabilisation loop

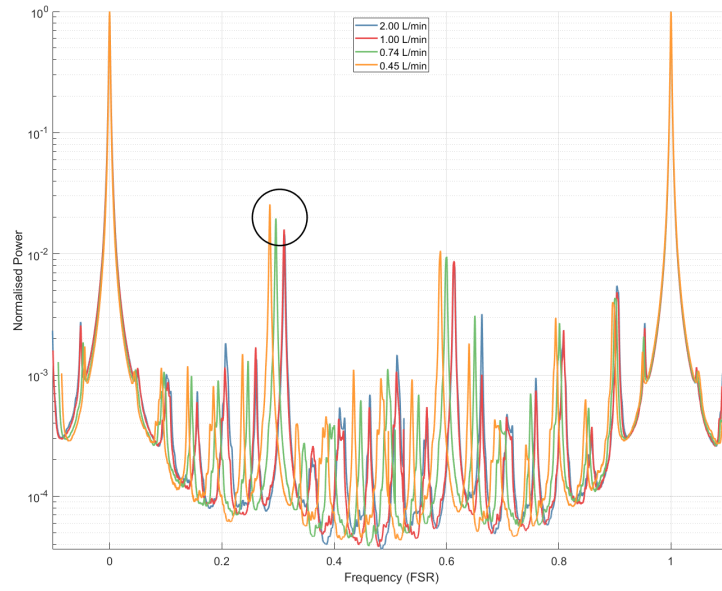


Figure 2.3: Flow rates affecting the thermal lensing, resulting in increased ellipticity

Chapter 3

Amplitude Stabilisation for High Power Laser at LIGO

3.1 Introduction + Requirements

- shot noise requirement
- radiation pressure noise setting noise requirement
- inner+outer loop design
- could include calculation for mismatches in the IFO? Point absorbers etc.

3.2 Free Running Noise and Noise Hunting

- free running noise of the laser
- what dominates the laser noise? NPRO
- noise sources such as mounts
- noise from PMC offset not being tuned
- 4Hz noise from Beckhoff

3.3 AOM Stabilisation

- description of AOM
- AOM TF
- AOM loop TF
- plot of AOM stabilisation working

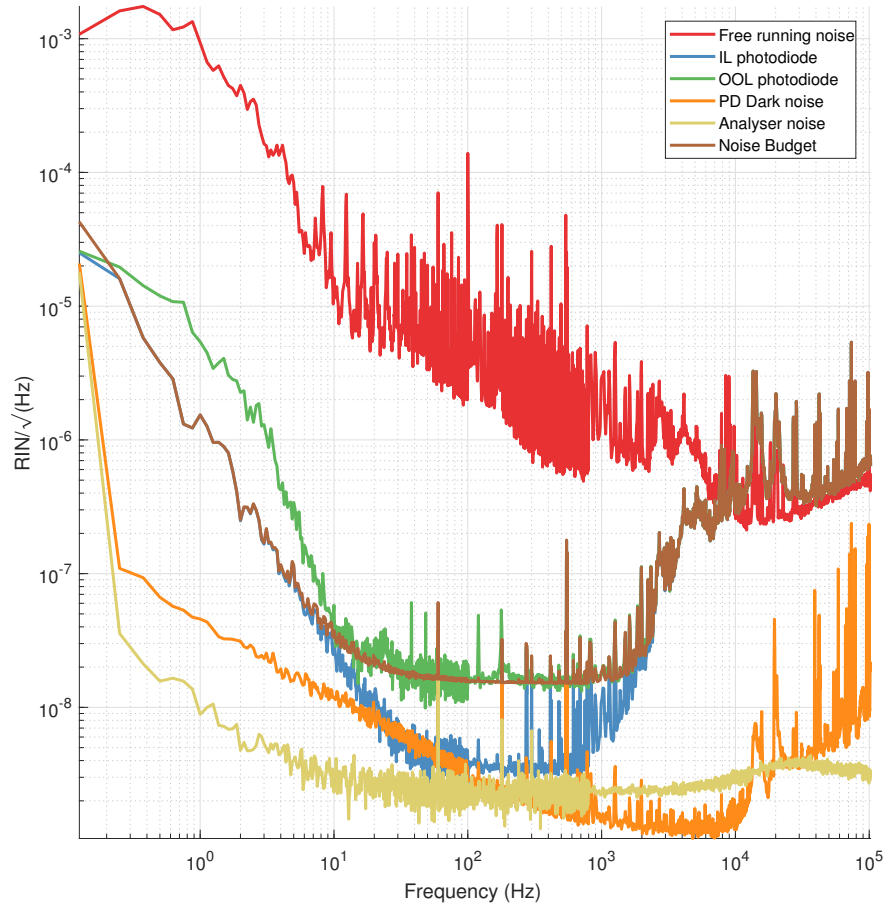


Figure 3.1: Free running and stabilised noise of laser

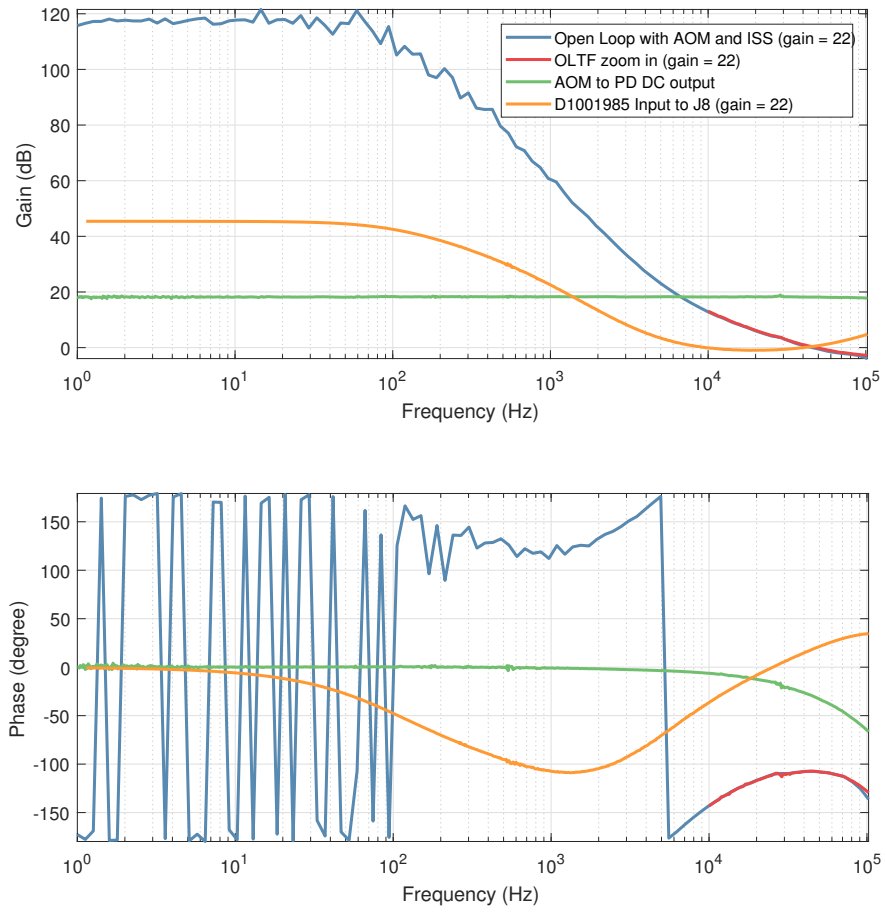


Figure 3.2: Transfer functions for parts of the AOM servo

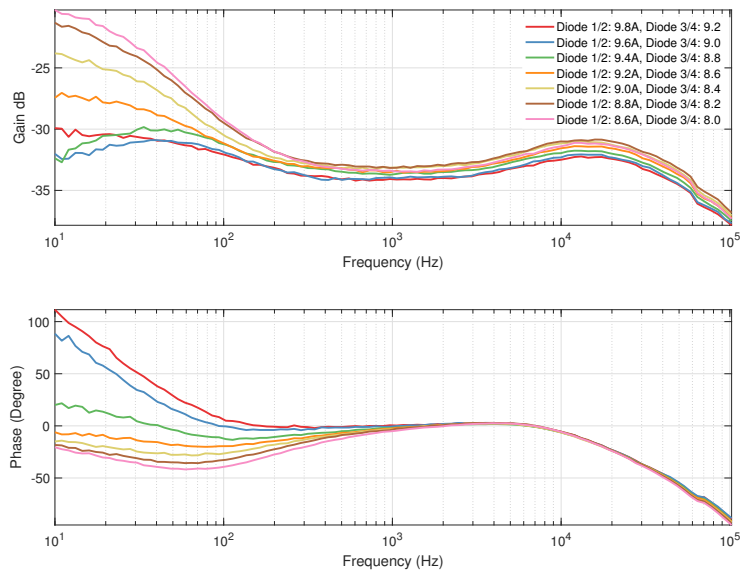


Figure 3.3: current shunt input to light power transfer function for different diode currents

3.4 Current Shunt Stabilisation

- description of CS
- CS TF
- CS loop TF
- plot of CS stabilisation working

3.5 Comparison of Two stabilisation Schemes

- why I think the AOM is a better choice even though they both work

3.6 Conclusion

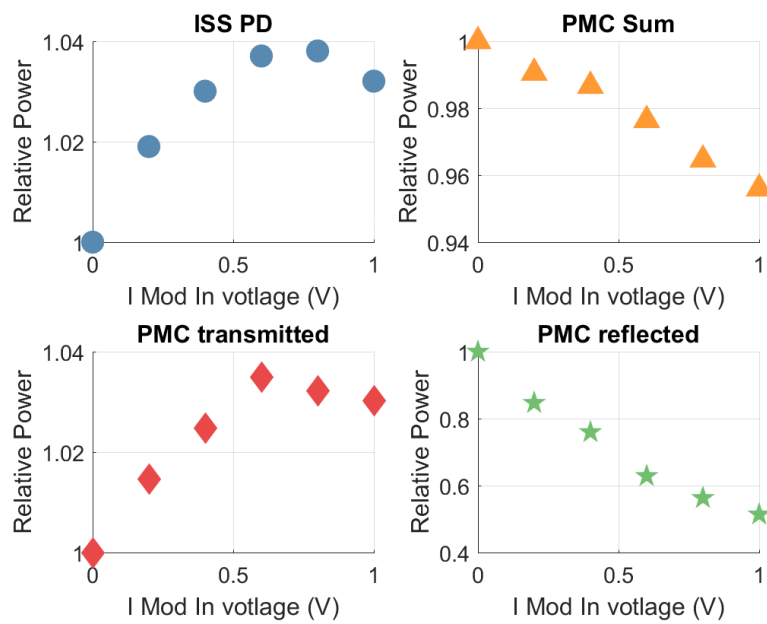


Figure 3.4: DC behaviour of current shunt

Chapter 4

Balanced Homodyne for LIGO A+

- 4.1 Motivation for BHD in A+
- 4.2 Derivation of path length stability
- 4.3 WS visualisations as a tool for understanding mode matching problems
- 4.4 Mode matching with astigmatism
- 4.5 Algorithm for calculating optimal 3 lens mode matching solution
- 4.6 HAM6 layout
- 4.7 Thermal lensing
- 4.8 Arm cavity power linked to AWC
- 4.9 Conclusion about BHD for A+

Chapter 5

in Air BHD experiments

This chapter is pedagogical in style, where the MZ is used to demonstrate shot noise being related to the wavelength of light. The in air MZ had unequal arm powers to create a signal LO interference style signal. From this chapter, will show shot noise limited measurement in terms of metres, and discuss various noise sources in the lab. By introducing a peak . The experiment was modelled, and there is a calculation based analysis as well.

Thoughts on 50:50ness of beam splitter

- total light the same regardless of splitting ratio
- contrast of MZ worse with unequal splitting ratio
- worse contrast = lower signal
- same total light
- so SNR lower
- Potential circuit improvement for BHD, better electronic noise potential from higher TIA gain

5.1 Introduction

5.2 Theory of MZ related to BHD

5.2.1 Link 50:50 ness of beam splitter to SNR

5.2.2 Link photodiode loss to SNR

5.2.3 Link LO LIGO phase noise to this experiment

5.2.4 Could think about loss at the beamsplitter, havent done that before

5.3 Apparatus

5.3.1 CDS and whitening/dewhitening

5.3.2 Servo

5.4 Measurement Procedure

explanation of how the apparatus was set up to take the measurement.

5.4.1 Calibration

Calibration of MZ to get from counts to metres:

<https://arran.physics.gla.ac.uk/wp/qnd-readout/2019/03/08/changed-injection-signal-to-mz>

Calibration of arm powers can be obtained from here (since relationship between contrast and arm power difference is known):

<https://arran.physics.gla.ac.uk/wp/qnd-readout/2019/02/26/re-alignment-of-mach-zehnder-a>

ND filter note <https://arran.physics.gla.ac.uk/wp/qnd-readout/2019/02/13/mz-nd-filter-insta>

5.5 Results and analysis

5.6 Conclusion

Chapter 6

In-vacuum experiments

6.1 Introduction

The suspended one looked at misalignments effect on signal, as well as being a more interesting control problem.

- made a suspended MZ in vacuum to get shot noise at lower frequencies and have control over angle of mirrors (angular dither)
- control theory important here
- investigated amplitude modulation
- investigated frequency noise with this as well (with frequency modulated laser)

6.2 apparatus

6.2.1 description of vacuum system

6.2.2 suspensions

6.2.3 photodiodes

6.2.4 measurement of table motion

6.3 control loop

6.3.1 transfer function simulation of suspension

6.3.2 analogue electronics

6.3.3 digital filtering

6.4 Measurements

6.5 Discussion

Chapter 7

Characterisation of InGaAs Photodiodes

should this be an appendix?

Chapter 8

Characterisation of 2um Photodiodes

8.1 Introduction and requirements

8.1.1 Wavelength choice

8.1.2 Semiconductor choice

8.1.3 Quantum efficiency

8.1.4 linearity

8.1.5 Quantum noise from modulation demodulation emphasis

8.1.6 low levels of dark noise ($1/f$)

8.1.7 size of diode

8.2 ‘Bare minimum’ Photodiode Theory

8.2.1 what is a semiconductor

8.2.2 what is a diode

8.2.3 what is a PIN junction

8.2.4 what is a photodiode and basic properties of them

8.2.5 Possible materials for photodiodes

8.2.6 Dark current

8.2.7 $1/f$ noise sources in semiconductors

8.2.8 growth techniques for photodiodes and why its important

8.3 List of experiments done on extended InGaAs

to test the requirements above...

8.3.1 dark noise measurements

dark noise as function of bias

dark noise as function of temperature

8.3.2 Light based tests

lock-in style measurements to measure QE change

DC style measurements to measure QE change

Shot noise measurement to validate QE change

lock in style measurement to test linearity

8.3.3 List of experiments done on Hammamatsu P5968-100 InSb

Promising data sheet, showed high QE and linearity. This is still to be done

8.4 Apparatus for measurements in each subsection above

8.5 Discussion of the results for each subsubsection above

8.6 Conclusion

8.7 Current write up in paper form

Below is currently just the paper I was working on. More detail will go in, as well as additional measurements + results which I haven't written up. This is here a place holder for now.

8.8 abstract

High quantum efficiency, low noise photodiodes are a fundamental part of the readout of ground based gravitational wave detectors. The 3rd generation of these detectors may be operated with either a 1550 nm or around 2000 nm wavelength laser as a consequence for the potential to develop cryogenic compatible coatings and test masses for these wavelengths. The current InGaAs photodiodes which are used for the readout of gravitational wave detectors would not be compatible with a 2 μm laser. Extended InGaAs is a potential material that could be used for these photodiodes and can be purchased with a variety of cut-off wavelengths up to 2.6 μm . In this paper, several off-the-shelf extended InGaAs photodiodes with different cut-off wavelengths were characterised. The dark noise, linearity and quantum

efficiency of these photodiodes was investigated to see if there were optimal operating conditions which would provide a high enough quantum efficiency while still being low enough noise as to not limit the noise performance of the interferometer. It was found that the photodiodes tested do not meet the requirements of a 3rd generation gravitational wave detector. Longer cut-off wavelength photodiodes tended to perform more poorly than photodiodes with a shorter cut off wavelength in terms of noise performance and linearity, although this varies between manufacturer. While under a usable reverse bias voltage the highest quantum efficiency measured was 70%. While its possible to achieve higher quantum efficiencies the noise of the photo-current will not be limited to shot noise in the detection frequency band of a gravitational wave detector, ~ 100 Hz, due to large amounts of $1/f$ noise. This due to it being technically challenging to fabricate extended InGaAs which is thick enough to have a high quantum efficiency while also having a low amount of defects.

8.9 Introduction

Einstein predicted the existence of gravitational waves, and since 2015 a catalogue of detections has been made [1]. As well as being a test of general relativity, the detection of gravitational waves has ushered a new era of astronomy, giving us the ability to directly observe astrophysical objects such as binary black holes and binary neutron stars [2]. With a global network of kilometre scale interferometers, multi-messenger astronomy has already yielded exciting new results. Measurements of gravitational waves emitting the from merging of binary neutron stars has tested the equivalence principle [3], given us bounds on the speed of gravity and allowed us to test the value of the Hubble constant [4], as well as giving us insights into the origin of heavy elements [5].

As the current generation of gravitational wave detectors reach their design sensitivity, researchers are looking to the future and how to improve the sensitivity of the next generation of detectors. Once at design sensitivity, the noise floor of LIGO will be limited by mirror coating thermal noise (Brownian motion of the molecules in the mirror surface) in the detection band (around 100 Hz)[6]. Since these thermally driven fluctuations are a limiting source of noise [7], proposed future gravitational wave detectors such as the Einstein Telescope [8] and LIGO Cosmic Explorer [9] will use cryogenically cooled mirrors. To take full advantage of this approach will require the development of new cryogenically optimised coatings and switching the operating laser wavelength to 1550 nm or around 2000 nm, instead of the 1064 nm wavelength used in current gravitational wave detectors. Changing the wavelength of the laser to 2 μ m means that the current InGaAs photodiodes used in the readout of gravitational wave detectors will not be suitable since typical InGaAs photodiodes are not sensitive to 2 μ m light.

Current gravitational wave detectors operate in the 1-10 mW range at the output [10] to ensure that the junk light from other modes is a small portion of the total signal. Thus, any new photodiode should be able to produce ~ 10 mA of photo-current. The photodiodes used in the readout of a gravitational wave interferometer should have a quantum efficiency of more than 99%. The reason for having such a high quantum efficiency requirement is that to

achieve the baseline level of ~ 10 dB of squeezed light that future gravitational wave detectors will use, losses in the interferometer have to be extremely low as losses degrade the amount of squeezing achieved. Photodiodes for gravitational wave detectors should have low $1/f$ dark noise so that they are not limited in the detection band by electronic noise. Third generation ground based gravitational wave detectors will be quantum noise limited down to about 10 Hz [11], so the $1/f$ noise must be below the shot noise of the photo-current above this frequency. The $1/f$ noise could potentially be reduced by using external modulation, however the shot noise would increase if the mixing of the modulation and demodulation waveforms was not time independent [12], which limits the choice of waveforms. A modulation-demodulation scheme such as external modulation with a π square wave has been considered, but this comes with a significant technical difficulty in creating a suitable EOM driver, and remains to be proven [12]. Even if the EOM was driven with a good enough square wave, the use of squeezing means that this approach would still not be sufficient as adding higher harmonics boosts the signal rather than reduces noise. In other words, the noise improvement from squeezing would be lost.

If the wavelength that gravitational wave detectors are operated at changes to $2\text{ }\mu\text{m}$ then extended InGaAs would be a candidate material for the readout photodiodes. Extended InGaAs has more than the usual 53% indium found in conventional InGaAs diodes, as this reduces the size of the bandgap thus and increases the cut-off wavelength. Increasing the reverse bias voltage increases the quantum efficiency of extended InGaAs photodiodes, and data sheets for these devices show that there is a relationship between quantum efficiency and temperature. Several extended InGaAs photodiodes were tested to see if there was a combination of bias voltage and temperature that would improve the quantum efficiency whilst keeping the $1/f$ noise low. It was found that noise at temperatures between 0°C and room temperature was almost entirely dependant on reverse bias voltage and did not improve as the device was cooled. It was found that to get adequate noise performance, the diodes needed to be used at a reverse bias voltage that meant their quantum efficiency was far below the requirement of third generation gravitational wave detector. This $1/f$ noise and quantum efficiency problem has limited previous squeezing experiments using $2\text{ }\mu\text{m}$ lasers [13]. For instance, the FD10D from Thorlabs was measured to have a quantum efficiency of 82% at 1550 nm when used at its full rated reverse bias, and at this bias the $1/f$ noise only became less than a typical shot noise above 1 kHz. When asked about the possibility of procuring a diode that meets the requirements set out above, the responses from manufacturers ranged from saying such a diode is impossible to construct to saying that money is the main obstacle. PIN photodiodes use a positive-intrinsic-negative semiconductor structure to create a device that when light of sufficient energy, energy greater than the band gap of the semiconductor, is incident on it an electron-hole pair is produced. Within the P region there is an excess of holes and within the N region there is an excess of electrons. This region is called the depletion region. The intrinsic semiconductor is used to make the depletion region of the photodiode wider. Due to the electric field between the P and N regions, electron-hole pairs that are produced in the depletion region due to an incident photon results in electrons moving towards the N-type semiconductor and holes drifting towards the P-type semiconductor, thus creating a current. By applying a reverse bias,

the depletion region is widened resulting in the conversion of light to current being more efficient. The ratio of total photons incident on the photodiode to photons which produce electrons-hole pairs is called quantum efficiency.

Photodiodes produce a dark current I_d which depends on the reverse bias. The dominant source of dark current depends on the regime of reverse bias that the diode is under and can be separated into three sources: diffusion current I_{diff} ; generation-recombination current I_{GR} and quantum tunnelling current I_{tun} [14],

$$I_d = I_{\text{tun}} + I_{\text{diff}} + I_{\text{GR}}. \quad (8.1)$$

I_{tun} arises due to electrons being able to tunnel across the band gap, similar to the breakdown voltage of a diode. I_{diff} arises due to the diffusion of minority carriers into the intrinsic region. I_{GR} is due to the generation and recombination of electron hole pairs in the intrinsic region, and the random nature of the generation and recombination of electron hole pairs is the source of $1/f$ noise [15]. This noise is sometimes referred to as Shockley-Read-Hall (SRH) noise (or more generally for $1/f$ noise with a different physical origin as random telegraph or popcorn noise). Electrons and holes enter what are known as trap states, states which ideally would not exist, and the amount of time spent there before leaving and recombining with electrons and holes is random. These states can arise due to lattice mismatching. The spectrum of one trap causing generation-recombination of electrons and holes is a Lorentzian which when many traps are near each other can look like a $1/f$ spectrum [16][17][18]. This theory also allows for noise spectra which are nearly $1/f$, or have bends in them. InGaAs is a widely used semiconductor for fabricating photodiodes. Typically they are sensitive up to wavelengths around 1550 nm, however longer wavelength sensitivity can be achieved by increasing the amount of indium doping. These photodiodes are often called ‘extended’ and can be sensitive for wavelengths up to 2.6 μm . One of the main issues faced when creating an extended InGaAs photodiode is the lattice mismatch between the InGaAs and the InP substrate on which its grown. Lattice mismatching can arise due to a strained crystal structure, and since the extra indium in an extended InGaAs photodiode causes this [19] it is hard to fabricate a low noise extended InGaAs photodiode [20]. Above 53% indium, a buffer layer of InGaAs with less indium in between the absorbing layer and the substrate can be used to mitigate the lattice mismatch slightly, but the dislocations still give rise to trap states which cause the generation and recombination of charge carriers [21]. Lattice mismatching is the source of the large amounts of dark current found in extended InGaAs as well as the $1/f$ noise.

Creating a photodiode with the quantum efficiency requirements of third generation gravitational wave detectors is difficult to achieve as the lattice mismatch makes it hard to make the photodiode thick enough since the critical thickness of extended InGaAs is only hundreds of angstroms [19]. The parameters that affect the internal quantum efficiency are the absorption coefficient of the extended InGaAs and the amount of the extended InGaAs material the light must travel through. For InGaAs that is sensitive to 2 μm the absorption coefficient, α , is approximately 10^4 cm^{-1} [22]. To estimate the required thickness, x , of InGaAs to achieve 99% quantum efficiency, we can use the Beer-Lambert law, $\frac{I_x}{I_0} = e^{-\alpha x}$ to find that

$x = 4.6 \mu\text{m}$. Although this is just a rough calculation, it does give one an idea of how thick the material needs to be. This assumes all the photons absorbed make it out of the junction as electrons/holes, which may not be the case as without a high enough bias the electrons will fail make it out of the junction before recombining with a hole.

To ensure a low amount of strain, the layer of InGaAs could only be about 100 nm thick between the InP and the InGaAs layer if the diode were to be constructed in the typical PIN arrangement. Above this thickness, dislocations due to the InGaAs not being lattice matched to the InP are a problem. The thicker the InGaAs layer is, the more strain relief is required between the InP substrate and the InGaAs layer. The more strained the InGaAs is, the more dislocations there will be which will lead to more $1/f$ noise and dark current. The critical thickness is the thickness above which the semiconductors quality degrades significantly, and the higher the ratio of indium to gallium, the thinner this critical thickness becomes [23]. Putting 100 nm into the Beer-Lambert law gives an internal quantum efficiency of about 10% which is far below the requirements for GW detectors. The amount of threading dislocations contained within an InGaAs photodiode can be reduced by orders of magnitude with careful choice of growth conditions and design of the layers in the photodiode [24][21].

Another design challenge arises because InGaAs is not a good conductor of heat so a thicker amount of InGaAs can lead to lower power handling before the diode is thermally destroyed[25]. Once the internal quantum efficiency problem has been solved, an anti reflection coating would need to be developed. InGaAs that is optimised for $2 \mu\text{m}$ has a refractive index, n , of approximately 3.6 [22]. Assuming a single quarter-wave layer was to be applied, the film would need a refractive index that was $n_f = \sqrt{3.6}$ and be 600 nm thick. The idea of using this anti reflective coating as a passivation layer is presented in [26].

8.10 Apparatus

When designing a photodiode circuit, the parameter space includes (but isn't limited too): reverse bias voltage used, size of the diode, temperature of the diode, cut-off wavelength and vendor. Each of these parameters could effect the noise performance and quantum efficiency of the photodiode, so experiments were designed to gain insights into where a diode which is suitable for a third generation gravitational wave detector readout scheme resides inside this space. The reverse bias changes the amount of $1/f$ dark noise of the photodiode. Therefore, measuring when the reverse bias would cause the dark noise to become larger than the shot noise for a typical light level in a gravitational wave detector was investigated. Different levels of Indium doping effects the amount of strain that the semiconductor experiences, so the dark noise of devices with different cut-off wavelengths was investigated. The size of the diode also can effect the amount of dark noise, since there is more material to generate the noise, so devices of different sizes were characterised. If certain vendors had better manufacturing processes, that would also affect the noise. The dependence of dark $1/f$ noise on temperature was investigated to see if this $1/f$ dark noise can be reduced if the temperature is decreased from room temperature to 0°C . To gain insights into how each of the variables mentioned affect the dark noise and to have a

Part Name	Vendor	Cut-off Wavelength	diameter
IG19x1000S4i4	Laser Components	1.9 μm	1 mm
IG22x1000S4i4	Laser Components	2.2 μm	1 mm
IG24x500S4i4	Laser Components	2.4 μm	0.5 mm
IG26x500S4i4	Laser Components	2.6 μm	0.5 mm
FD01D	Thorlabs	2.6 μm	1 mm
G12183-020K	Hamamatsu	1.9 μm	2 mm

Table 8.1: Extended InGaAs photodiodes used in the experiment.

workable size of the parameter space to explore, the diodes shown in table 8.1 were tested. To help identify the mechanism by which the dark noise is created in these kinds of diodes, the forward current voltage curve was measured and the ideality factor was determined.

The requirements for an experimental characterisation of the chosen photodiodes are a) a dark box within which the experiment may be conducted, b) availability of a variable, low-noise bias voltage and c) a means by which the temperature of the device under test can be varied and controlled. Further, a suitably low-noise amplifier is required to monitor the photo-current and convert it to a voltage. To measure the dark current produced by multiple different photodiodes under different reverse bias voltages and easily obtainable temperatures, a dark box containing a Peltier module was constructed. The photodiode was held in place against a piece of aluminium that was also in contact with the Peltier and an LM35 temperature sensor was used to measure the temperature of the aluminium. The dark current of the photodiodes was expected to be of the order nanoamps so a 1 M Ω resistor was used in the transimpedance amplifier. Due to this choice of resistor a TL071 op amp was used, as shown in Figure 8.1. The bias voltage was provided by a LM7815 voltage regulator and since not much current was being drawn a variable resistor was used to set the bias voltage. A 1000 μF electrolytic capacitor was used to reduce the bias voltage noise. The low noise fixed bias circuit shown in figure 8.7 was constructed to verify that the more flexible swappable photodiode dark noise circuit would work.

The quantum efficiency of the photodiode was measured as the reverse bias was changed. The reverse bias voltage affects the quantum efficiency since it affects the amount of time charge carriers spend in the depletion region, a lower bias allows more time for electrons and holes to recombine thus lowering the quantum efficiency. By measuring the photocurrent as the bias was increased for a fixed light power, the relative change in quantum efficiency as a function of reverse bias can be investigated. This can be verified as a genuine increase in quantum efficiency, and not a multi-charge carrier process, if the noise of the photocurrent can be stabilised to shot noise. Measuring the DC photocurrent and the amplitude of a modulation of the laser intensity as a function of reverse bias and light power allows for the region in which the photocurrent was linear to be determined. This requires an amplitude stabilised laser with a wavelength which the diodes in table 8.1 are sensitive to and a variable bias voltage circuit for testing the extended InGaAs photodiodes under different biases.

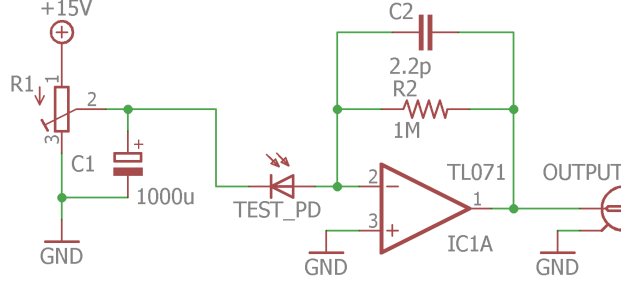


Figure 8.1: The transimpedance amplifier used to measure the dark noise of a variety of photodiodes. For large feedback resistors (R_2 in the circuit above), a JFET style OP amp such as the TL071 performs better since these have low current noise (e.g. see [27] for a description of noise sources in OP amp circuits). In this application, the TL071 is optimised for between $100\text{ k}\Omega$ to $10\text{ M}\Omega$.

To measure the relative quantum efficiency and linearity of photodiodes at light powers up to 20 mW , a 1500 nm wavelength laser (PPCL300 from Pure Photonics) was used. The transimpedance amplifier shown in Figure 8.2 was built to test the photodiodes under different bias voltages. A low passed bench top power supply was used to provide the DC bias voltage. Without filtering it was found that the noise of the power supply was the limiting source of noise when there was laser light on the extended InGaAs photodiodes, as the impedance of the photodiodes decreased when light was on the photodiode. To prove that the increase in photo-current was due to an increase in the reverse bias, the shot noise of the photo-current was measured. To do this an amplitude stabilised laser was required. A none extended Excelitas InGaAs photodiode (C30641GH) was used for this and it was assumed this diode behaved well (this range of photodiodes is widely used in LIGO). The apparatus is shown in Figure 8.3. The half waveplate and polarising beam splitter allow for different DC light level to be controlled. Since at least 50% of the light was always directed onto the photodiode used for the amplitude stabilisation, the light was shot noise limited for all accessible light powers. The lens was used to adjust the spot size. To make the measurements, an spectrum analyser was used.

8.11 Experimental Results from Testing Extended In-GaAs Photodiodes.

The FD10D is a 1 mm photodiode sold by Thorlabs, with a cut-off wavelength of $2.6\text{ }\mu\text{m}$. Figure 8.6 shows why one would want to increase the bias as much as possible, as the amount of photo-current generated increases by 17% for this particular diode when the reverse bias is turned up to its maximum data-sheet value [28]. The measured dark noise for this photodiode (see Figure 8.4) shows that running this diode with a reverse bias much higher than 0.65 V would result in $1/f$ noise larger than the shot noise of a 10 mA photo-current at the frequencies of interest ($\sim 100\text{ Hz}$). If used as a gravitational wave detector readout photodiode, the sensitivity of the detector would be limited by $1/f$ technical noise rather than quantum-limited noise floor of shot noise. The absolute quantum efficiency was

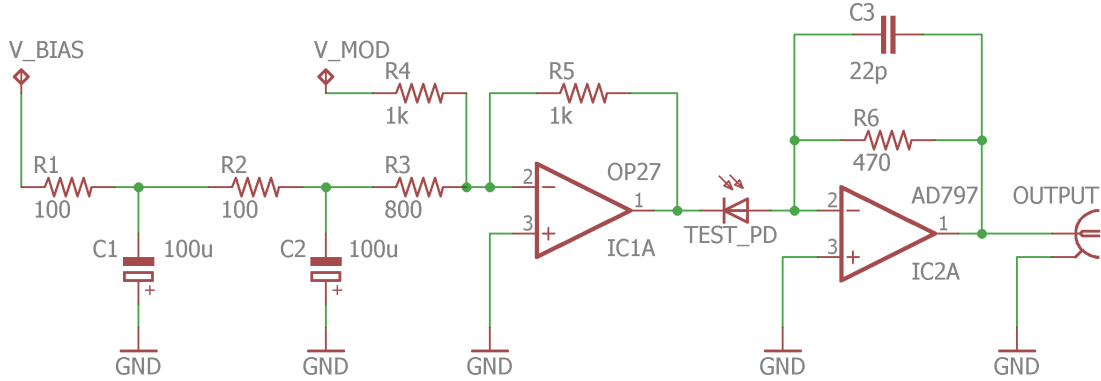


Figure 8.2: Transimpedance amplifier used to measure the quantum efficiency and linearity of photodiodes for up to 20 mW of 1550 nm laser light. The reverse bias supplied to the photodiode is provided by the first OP amp (OP27). The two inputs V_{bias} and V_{mod} allows for a quiet low passed DC bias to be set while also being able to perform lock-in type measurements.

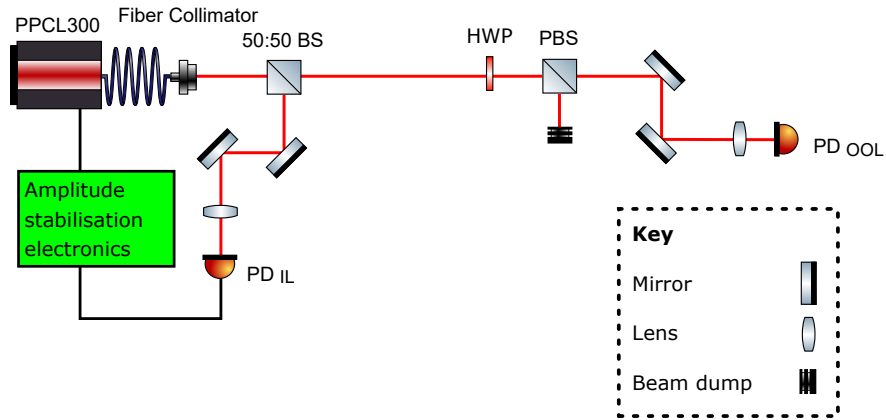


Figure 8.3: Layout of experiment. PD_{IL} is the in-loop photodiode (C30641gh) and PD_{OOL} is the photodiode being tested. The PPCL300 is a 1550 nm fibre laser from pure photonics. Since the C30641gh is a well-characterised diode, this was used as the in-loop diode for the amplitude stabilisation

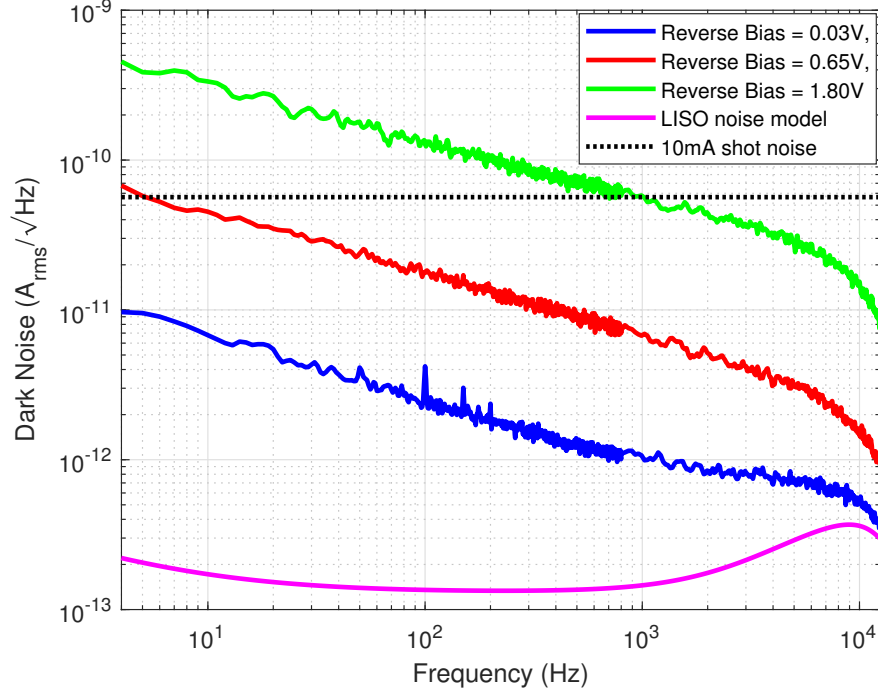


Figure 8.4: The dark noise of the FD10D at different reverse biases. The maximum datasheet value is 1.8V, above this voltage electrons can start tunnelling across the band gap. The rolling off above 9 kHz is an artefact of the circuit. Above 0.65 V reverse bias, the $1/f$ noise becomes larger than shot noise. At this reverse bias, the quantum efficiency is $\sim 70\%$. These measurements were made at room temperature.

measured for this diode using multiple power meters. Taking into account reflections from the glass window in front, the quantum efficiency was measured to be 82% (the power of the laser was cross checked with 2 different power meters, these agreed to within less than 0.5%) at 1.8 V reverse bias. At 0.6 V reverse bias the $1/f$ noise is below the target, but at the expense of the quantum efficiency which was measured to be 70%. The maximum photo-current this diode was tested at 0.6V reverse bias was 15 mA and this showed no sign of saturation. Although the FD10D did manage to handle a reasonable amount of power, the quantum efficiency was still far from the target of greater than 99%.

It was necessary to check that the power-supply/bias voltage noise was not be effecting the result. In order to show the swappable photodiode design met the noise requirements, a fixed bias voltage with noise at 10Hz of $50 \text{ nV}_{\text{rms}}/\sqrt{\text{Hz}}$ was used to cross check the measurement of dark noise for the FD10D. This requirement was set by assuming that the dynamic impedance of the test photodiodes would be of the order 100 k Ω , and the smallest amount of current noise that is measurable at the inverting input of the TL071 in this circuit is $0.5 \text{ pA}_{\text{rms}}/\sqrt{\text{Hz}}$, so multiplying these two quantities gives the bias noise requirement. By having a socket rather than a solder joint and a standard voltage regulator to provide the bias, there is potential to have bias voltage noise of several 100 $\text{uV}_{\text{rms}}/\sqrt{\text{Hz}}$. A low noise reference voltage (AD587) with 2 low pass filters was used to create the quiet bias voltage (see Figure 8.7).

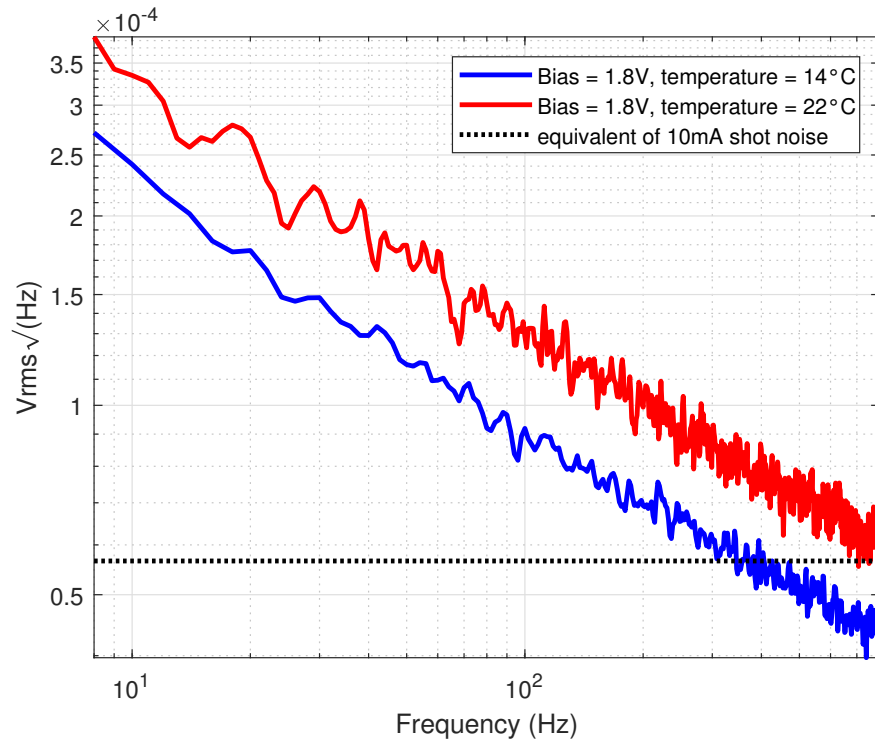


Figure 8.5: The dark noise of the FD10D with a reverse bias of 1.8V at room temperature and 14°C. Although there is a slight reduction in noise, this comes at the expense of less quantum efficiency near the cut off wavelength.

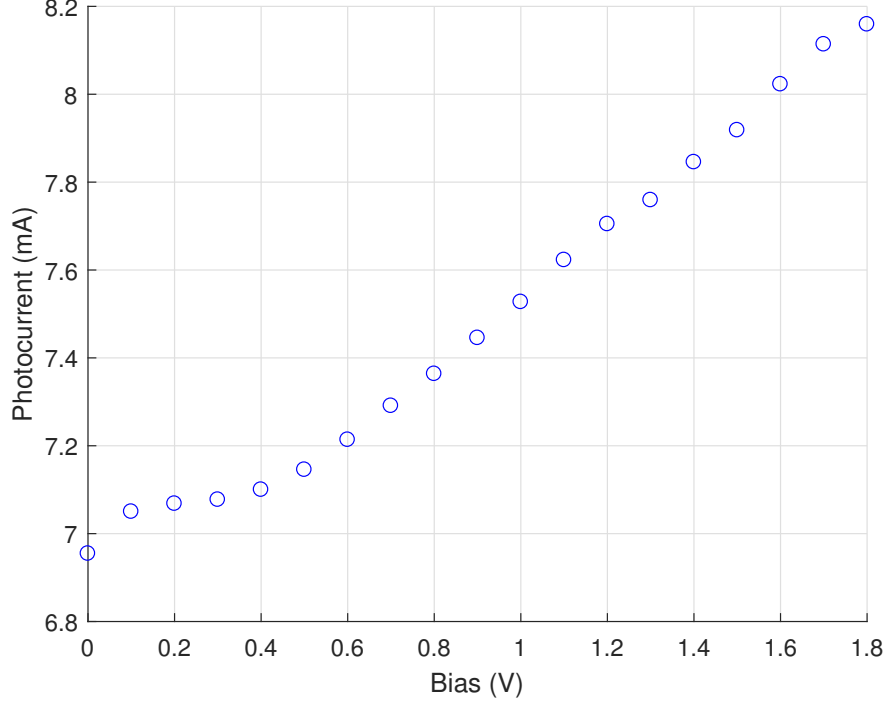


Figure 8.6: A 1550nm laser was used to illuminate the FD10D at room temperature. As the reverse bias is increased, the photo-current out of the FD10D increases by about 17%. This increase in photo-current is due to an increase quantum efficiency since the noise of photo-current matched shot noise.

There was no difference in noise between using the swappable photodiode test circuit and the quiet bias voltage circuit (see Figure 8.8), so we can rule out bias voltage noise being the cause of the $1/f$ noise. If the bias voltage produced $1/f$ noise and the photodiode also produced $1/f$ noise, then you would see a corner in the spectrum, as well as a generally messy looking spectrum.

The large amounts of $1/f$ noise was not unique to the FD10D - several extended InGaAs photodiodes from Laser Components were also tested. The main problem was the $1/f$ noise becoming large whenever the bias voltage was increased in an attempt to improve the quantum efficiency, necessitating the use a small reverse bias. A small reverse bias results in electrons-hole pairs failing to make it out of the junction before recombining, leading to a smaller quantum efficiency as well as causing the photodiode to saturate. The Laser Components extended InGaAs photodiodes that were tested performed poorly in both noise and power handling terms (see Figures 8.9 and 8.10).

To investigate the physical mechanism of the $1/f$ noise found in extended InGaAs photodiodes, the forward current voltage curve of the IG24x500S4i was investigated. The ideality factor, n , of a diode indicates the process by which the generation-recombination current is created. The ideality factor of the IG24x500S4i was measured using the apparatus shown in

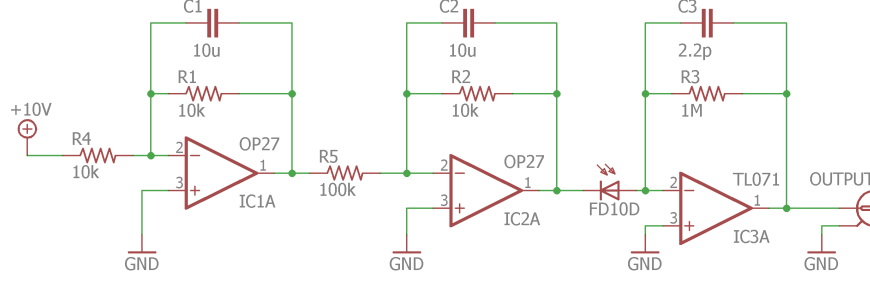


Figure 8.7: Low noise fixed bias voltage dark noise measurement circuit. The 10V reference is generated by an AD587, in this circuit the noise is limited by the voltage noise of the second OP27, but it still reaches $5 \text{ nV}_{\text{rms}}/\sqrt{\text{Hz}}$ at 10 Hz which is 10 times smaller than the amount of voltage noise which could be converted to current noise that would cause the signal at the output of the TL071 not to be limited by the current noise of the TL071, given the typical dynamic impedance for extended InGaAs photodiodes.

Figure 8.1 and compared to the model

$$I = I_s(e^{-qV/nk_B T} - 1) \quad (8.2)$$

where I_s is the saturation current, q is the electron charge, V is the voltage across the diode, k_B is the Boltzmann constant and T is the temperature. An ideality factor of 1 suggests that the generation-recombination current is driven by the existence of trap states (rather than a multi electron-hole process). This is shown in Figure 8.11. Fitting for I_s and n gives the saturation current to be 0.36 mA and the ideality factor to be 1.02.

8.12 Conclusion

Current off-the-shelf extended InGaAs photodiodes do not meet the requirements for readout schemes of future gravitational wave detectors given that simple PIN extended InGaAs photodiode diodes at room temperature are either too noisy, cannot handle enough power or do not have a high enough quantum efficiency. To overcome $1/f$ noise either a low bias is required which reduces the quantum efficiency, or a sophisticated modulation-demodulation scheme compatible with squeezing is required. A lower bias results in photons not being detected which detracts the amount of squeezing one can achieve, as does non-optimal chopping. With careful design of the layers within the photodiode and choice of fabrication process, the strain induced threading dislocations can be reduced, although suppliers of extended InGaAs photodiodes do not offer much hope when asked if a photodiode that meets the requirement of a third generation gravitational wave detector can be produced. The decision to run third generation detectors at $2 \mu\text{m}$ relies on being able to get high quantum efficiency, low noise photodiodes. Non-extended InGaAs photodiodes such as the ones used by LIGO would be compatible with using a 1550 nm laser. Other photodiode materials that are sensitive to $2 \mu\text{m}$ such as InSb and HgCdTe should also be investigated in terms of use for the read out of a gravitational wave detector.

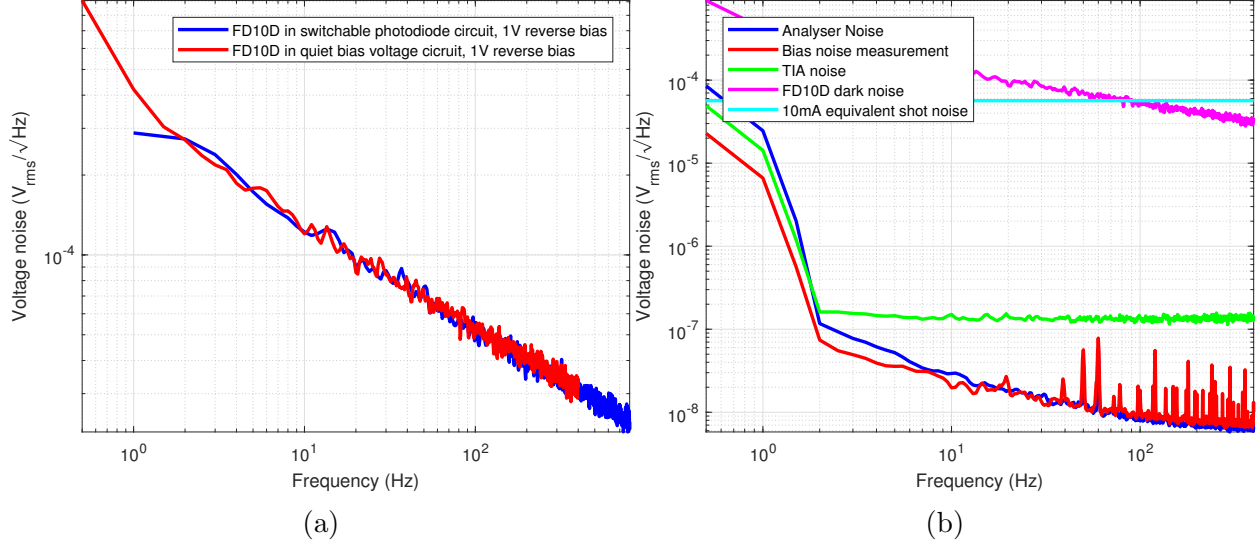


Figure 8.8: The left panel shows that the quiet reverse bias voltage source, shown in figure 8.7, and the more noisy but more flexible bias voltage source, shown in figure 8.1, gave the same results when the dark noise of the FD10D was measured. This means that the flexible reverse bias source is suitable to use for measuring the dark noise of other photodiodes. The bias voltage noise in the quiet bias voltage circuit was found to meet the requirements as shown in the right panel.

8.13 Acknowledgements

I would like to thank Andrew Long for our useful discussion about InGaAs, Sean Leavey for discussions about photodiodes, and Sebastian Steinlechner for providing us with some Laser Components photodiodes to test.

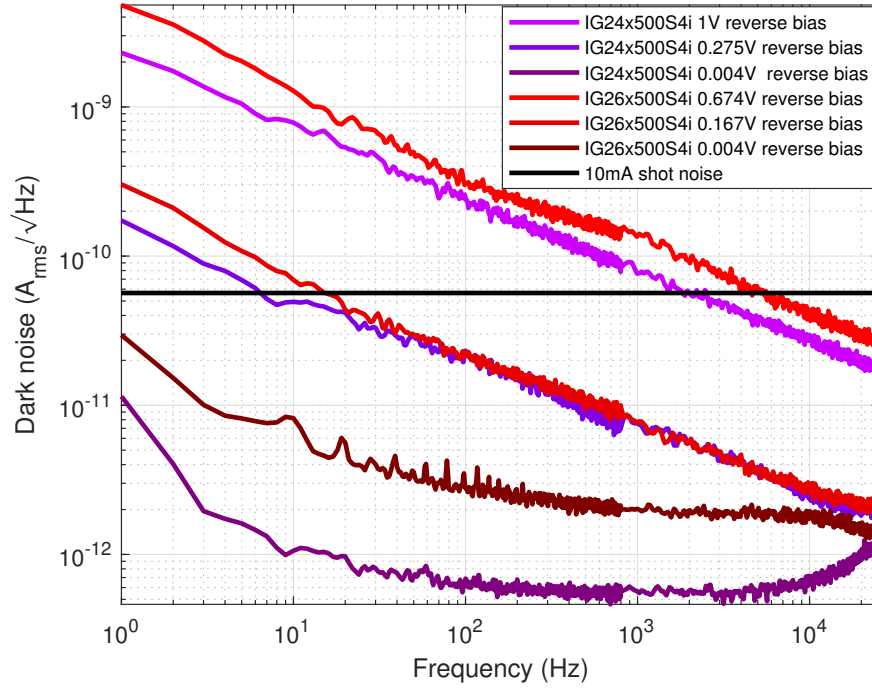


Figure 8.9: The dark noise of two different extended InGaAs photodiodes from Laser Components measured at room temperature. The noise curve for the IG24x500S4i at 0.004V is limited by the noise of the circuit shown in Figure 8.1. As the reverse bias is reduced, so is the $1/f$ noise. The IG26x500S4i had to be used at a lower reverse bias than the IG24x500S4i to achieve the same noise performance.

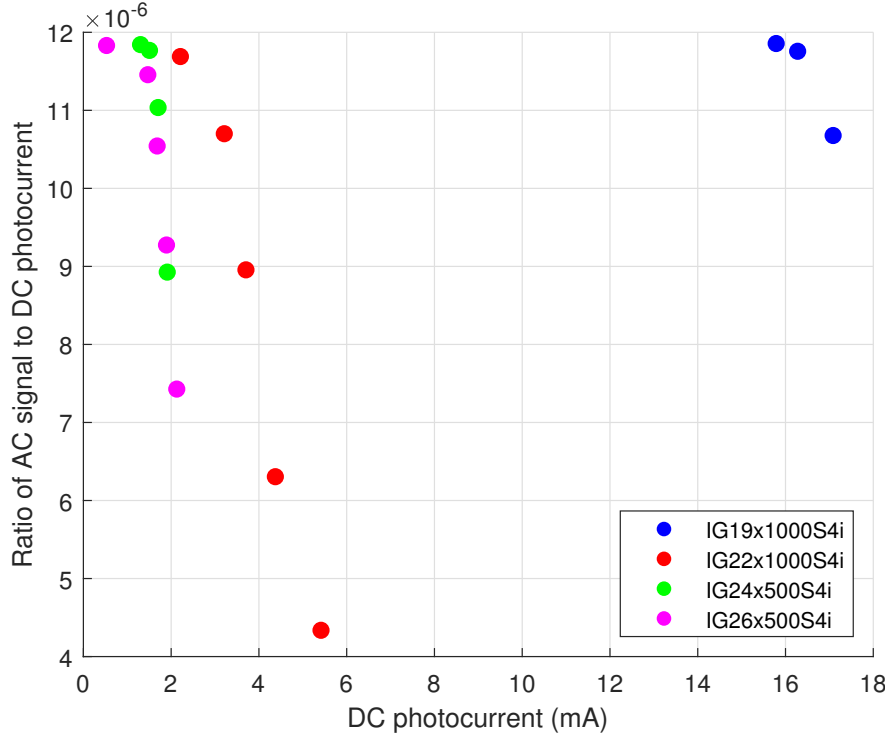


Figure 8.10: Using the apparatus shown in figure 8.3, the DC photo-current that the Laser Components extended InGaAs photodiodes could be operated with while still being linear (i.e. the AC component of the signal would scale linearly with the DC photo-current) was investigated. The spot size on the photodiodes was made as large as possible while still containing 99% of the light power on the photodiode, and were operated with the maximum reverse bias voltage that would still give $1/f$ noise less than 10mW shot noise at the frequencies of interest for the a gravitational wave detector. The laser's amplitude was modulated. Using a lock-in technique the AC response was measured and divided by the DC photo-current. This was compared to the AC/DC response of the in-loop regular InGaAs photodiode which behaved linearly. When the AC/DC ratio starts to become smaller, the photodiode is no longer behaving linearly. As these diodes become more extended, they perform more poorly.

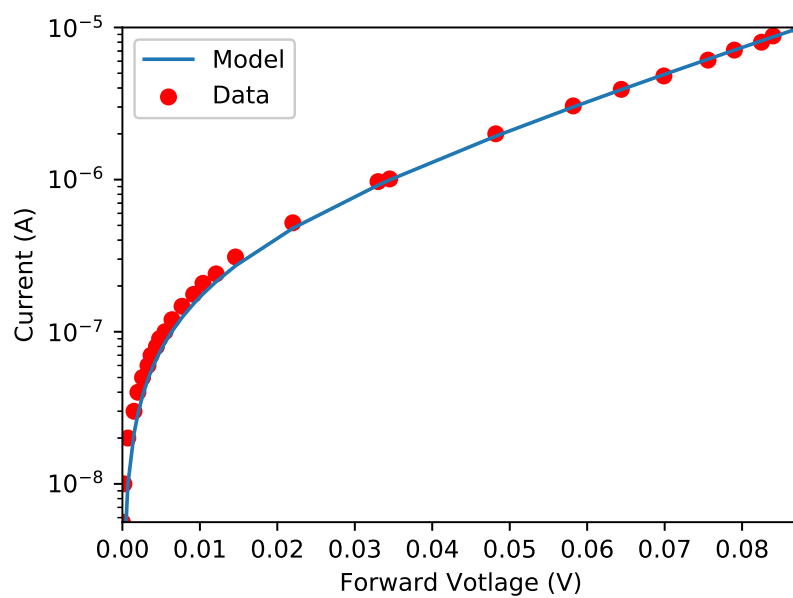


Figure 8.11: The forward IV curve of the IG24x500S4i.

Chapter 9

Conclusion and Future Work

Appendix A

Control Theory

Bibliography

- [1] The LIGO Scientific Collaboration and the Virgo Collaboration. Gwtc-1: A gravitational-wave transient catalog of compact binary mergers observed by ligo and virgo during the first and second observing runs.
- [2] LIGO Virgo Collaboration. Gw170817: Observation of gravitational waves from a binary neutron star inspiral. *Phys. Rev. Lett.*, 119:161101, Oct 2017.
- [3] LIGO Virgo Collaboration. Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817a. *The Astrophysical Journal*, 848(2):L13, oct 2017.
- [4] The LIGO Scientific Collaboration, The Dark Energy Camera GW-EM Collaboration The Virgo Collaboration, The 1M2H Collaboration, and The Las Cumbres Observatory Collaboration-The VINROUGE Collaboration & The MASTER Collaboration the DES Collaboration, The DLT40 Collaboration. A gravitational-wave standard siren measurement of the hubble constant. *Nature*, oct 2017.
- [5] Daniel Kasen, Brian Metzger, Jennifer Barnes, Eliot Quataert, and Enrico Ramirez-Ruiz. Origin of the heavy elements in binary neutron-star mergers from a gravitational-wave event. *Nature*, oct 2017.
- [6] L. Barsotti, L. McCuller, M. Evans, and P. Fritschel. The a+ design curve. LIGO DCC, April 2018. T1800042.
- [7] B. Lantz, S. L. Danilishin, S. Hild, Gustafson E., D. Coyne, V. Quetschke, G. D. Hammond, R. Adhikari, M. Evans, and R. Bassiri. Instrument science white paper 2018. LIGO DCC.
- [8] M Punturo, M Abernathy, F Acernese, B Allen, N Andersson, K Arun, F Barone, B Barr, M Barsuglia, M Beker, N Beveridge, S Birindelli, S Bose, L Bosi, S Braccini, C Bradaschia, T Bulik, E Calloni, G Cella, E Chassande Mottin, S Chelkowski, A Chincarini, J Clark, E Coccia, C Colacino, J Colas, A Cumming, L Cunningham, E Cuoco, S Danilishin, K Danzmann, G De Luca, R De Salvo, T Dent, R De Rosa, L Di Fiore, A Di Virgilio, M Doets, V Fafone, P Falferi, R Flaminio, J Franc, F Frasconi, A Freise, P Fulda, J Gair, G Gemme, A Gennai, A Giazotto, K Glampedakis, M Granata, H Grote, G Guidi, G Hammond, M Hannam, J Harms, D Heinert, M Hendry, I Heng, E Hennes, S Hild, J Hough, S Husa, S Huttner, G Jones, F Khalili, K Kokeyama,

- K Kokkotas, B Krishnan, M Lorenzini, H Lück, E Majorana, I Mandel, V Mandic, I Martin, C Michel, Y Minenkov, N Morgado, S Mosca, B Mours, H Müller–Ebhardt, P Murray, R Nawrodt, J Nelson, R Oshaughnessy, C D Ott, C Palomba, A Paoli, G Parguez, A Pasqualetti, R Passaquieti, D Passuello, L Pinard, R Poggiani, P Popolizio, M Prato, P Puppo, D Rabeling, P Rapagnani, J Read, T Regimbau, H Rehbein, S Reid, L Rez-zolla, F Ricci, F Richard, A Rocchi, S Rowan, A Rüdiger, B Sassolas, B Sathyaprakash, R Schnabel, C Schwarz, P Seidel, A Sintès, K Somiya, F Speirits, K Strain, S Strigin, P Sutton, S Tarabrin, A Thüring, J van den Brand, C van Leewen, M van Veggel, C van den Broeck, A Vecchio, J Veitch, F Vetrano, A Vicere, S Vyatchanin, B Willke, G Woan, P Wolfango, and K Yamamoto. The einstein telescope: a third-generation gravitational wave observatory. *Classical and Quantum Gravity*, 27(19):194002, sep 2010.
- [9] B P Abbott et al. Exploring the sensitivity of next generation gravitational wave detectors. *Classical and Quantum Gravity*, 34(4):044001, jan 2017.
- [10] M Prijatelj, J Degallaix, H Grote, J Leong, C Affeldt, S Hild, H Lück, J Slutsky, H Wittel, K Strain, and K Danzmann. The output mode cleaner of GEO 600. *Classical and Quantum Gravity*, 29(5):055009, feb 2012.
- [11] Gregory M Harry and the LIGO Scientific Collaboration. Advanced LIGO: the next generation of gravitational wave detectors. *Classical and Quantum Gravity*, 27(8):084006, apr 2010.
- [12] Brian J. Meers and Kenneth A. Strain. Modulation, signal, and quantum noise in interferometers. *Physical Review A*, 44(7):4693–4703, oct 1991.
- [13] Georgia L. Mansell, Terry G. McRae, Paul A. Altin, Min Jet Yap, Robert L. Ward, Bram J. J. Slagmolen, Daniel A. Shaddock, and David E. McClelland. Observation of squeezed light in the 2 μm region. *Physical Review Letters*, 120(20), may 2018.
- [14] S. Forrest. Performance of $\text{In}_x\text{Ga}_{1-x}\text{As}_y\text{P}_{1-y}$ photodiodes with dark current limited by diffusion, generation recombination, and tunneling. *IEEE Journal of Quantum Electronics*, 17(2):217–226, feb 1981.
- [15] Frank Seifert. *Power stabilization of high power lasers for second generation gravitational wave detectors*. PhD thesis, Hannover, Univ., Diss.; 2009, 2010.
- [16] Eric A.M. Klumperink, A.P. van der Wel, J.S. Kolhatkar, E. Hoekstra, Cora Salm, Hans Wallinga, and Bram Nauta. *Reduction of 1/f Noise by Switched Biasing: an Overview*, pages 307–315. Technology Foundation (STW), 11 2005.
- [17] Stefan Machlup. Noise in semiconductors: Spectrum of a two-parameter random signal. *Journal of Applied Physics*, 25(3):341–343, mar 1954.
- [18] F N Hooge, T G M Kleinpenning, and L K J Vandamme. Experimental studies on 1/f noise. *Reports on Progress in Physics*, 44(5):479–532, may 1981.

- [19] M. Gendry, V. Drouot, C. Santinelli, and G. Hollinger. Critical thicknesses of highly strained InGaAs layers grown on InP by molecular beam epitaxy. *Applied Physics Letters*, 60(18):2249–2251, may 1992.
- [20] Ramon U. Martinelli, Thomas J. Zamerowski, and Paul A. Longeway. 2.6 μm InGaAs photodiodes. *Applied Physics Letters*, 53(11):989–991, sep 1988.
- [21] Lars Zimmermann, Joachim John, Stefan Degroote, Gustaaf Borghs, Chris Van Hoof, and Stefan Nemeth. Extended wavelength InGaAs on GaAs using InAlAs buffer for back-side-illuminated short-wave infrared detectors. *Applied Physics Letters*, 82(17):2838–2840, apr 2003.
- [22] M. S. Alam, M. S. Rahman, M. R. Islam, A. G. Bhuiyan, and M. Yamada. Refractive index, absorption coefficient, and photoelastic constant: Key parameters of InGaAs material relevant to InGaAs-based device performance. In *2007 IEEE 19th International Conference on Indium Phosphide & Related Materials*. IEEE, may 2007.
- [23] Takayoshi Anan, Kenichi Nishi, and Sigeo Sugou. Critical layer thickness on (111)b-oriented InGaAs/GaAs heteroepitaxy. *Applied Physics Letters*, 60(25):3159–3161, jun 1992.
- [24] Xing you Chen, Yi Gu, and Yong gang Zhang. Epitaxy and device properties of InGaAs photodetectors with relatively high lattice mismatch. In *Epitaxy*. InTech, mar 2018.
- [25] Xiaowei Li, Ning Li, S. Demiguel, J.C. Campbell, D. Tulchinsky, and K.J. Williams. A comparison of front- and backside-illuminated high-saturation power partially depleted absorber photodetectors. *IEEE Journal of Quantum Electronics*, 40(9):1321–1325, sep 2004.
- [26] M. López, H. Hofer, K. D. Stock, J. C. Bermúdez, A. Schirmacher, F. Schneck, and S. Kück. Spectral reflectance and responsivity of ge- and InGaAs-photodiodes in the near-infrared: measurement and model. *Applied Optics*, 46(29):7337, oct 2007.
- [27] Analog Devices. *AD797 Ultralow Distortion, Ultralow Noise Op Amp*. Rev. K.
- [28] Thorlabs. *InGaAs Photodiode FD10D*, October 2017. Rev. D.